

Introduction

51% of burn wound related deaths are due to infection (Norbury et al, 2016). The need for safe/efficient technology limiting **infection** is critical. **Bromelain** is an FDA approved enzyme from pineapple used to treat burns with anti-inflammatory & antimicrobial properties. However, current applications rely on gauze and occlusive dressings and force patients to wait for hours – both **inconvenient** and wasteful in terms of supply and time.



Figure 1. Current bromelain cream treatment (Image taken from <https://www.nexobrid-us.com/>)

Background

Electrospinning creates nanofibers by shooting polymers with a high-voltage syringe. **Chitosan**, an antibacterial polymer, can be spun to form bandage-like sheets. **Hydrogels** (transparent, jelly-like) are another treatment with the ability to provide a proper microenvironment and promote tissue growth. Both treatments can be tested on **greater wax moth larvae**, a burn wound model that mirrors mammalian model results due to similar innate immune responses.

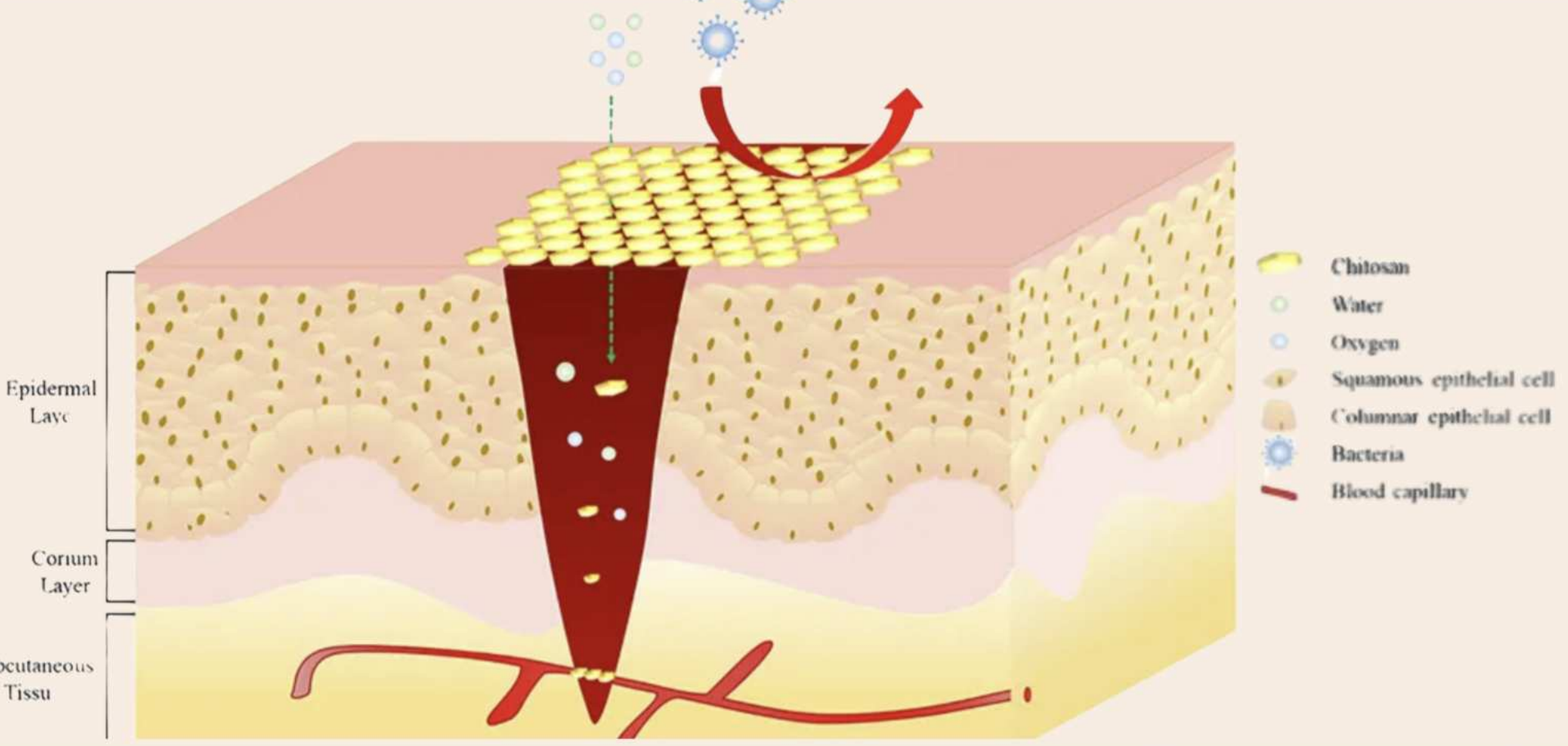


Figure 2. Chitosan's Wound Healing Properties (Xia et al., 2022)

Purpose

Engineering Goal

Determine the most effective method of **delivering bromelain** to limit infection of **Bacillus cereus**, a bacteria similar to that of common burn wound infectors. **Mortality** of larva will be measured to determine how well **treatments** perform in **treating infection**.

Hypothesis



Fiber's similar structure to the skin's **extracellular matrix** and bromelain's precedence of success for burn wounds will allow it to succeed to the greatest extent.

Novelty

Comparing **chitosan-based** treatments with bromelain in electrospun fibers and hydrogels and testing **electrospun fibers** on larva model.

Comparing Efficacy of Bromelain in Electrospun Fibers and Hydrogels for Burn Wound Treatment in the Invertebrate Burn Wound Model of *Galleria mellonella* Larvae

Category: Biomedical Engineering | Project #: 509-T-12

Methods

Testing Groups

Control Fiber (only chitosan)	Control Hydrogel (only chitosan)
Experimental Fiber (w/ bromelain)	Experimental Hydrogel (w/ bromelain)

No Treatment

Variables

IV – Presence of bromelain
DV – Mortality of larva

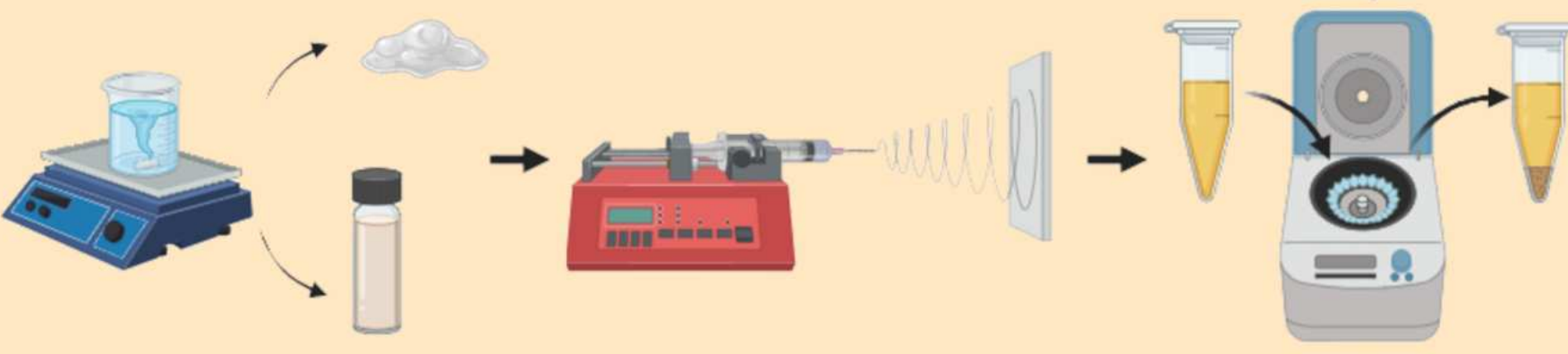


Figure 3. Procedure for Fiber and Hydrogel Preparation (Image created by Finalist using BioRender)

General Procedure

- Mix **composition chemicals** together
- Electrospin or grind ingredients (depending on treatment type)
- Create 10% bacteria solution

Fiber: PEO, Acetic Acid, and Chitosan
Hydrogel: Sodium Alginate, Acetic Acid, and Chitosan



Figure 4. Electrospinning apparatus with settings of 0.9mm diameter, 10 kV voltage, 10cm needle to collector distance, 0.5ml/h flow rate (Photo taken by Finalist)



Figure 5. Hydrogel in process of being made (Photo taken by Finalist)

PREP

TEST

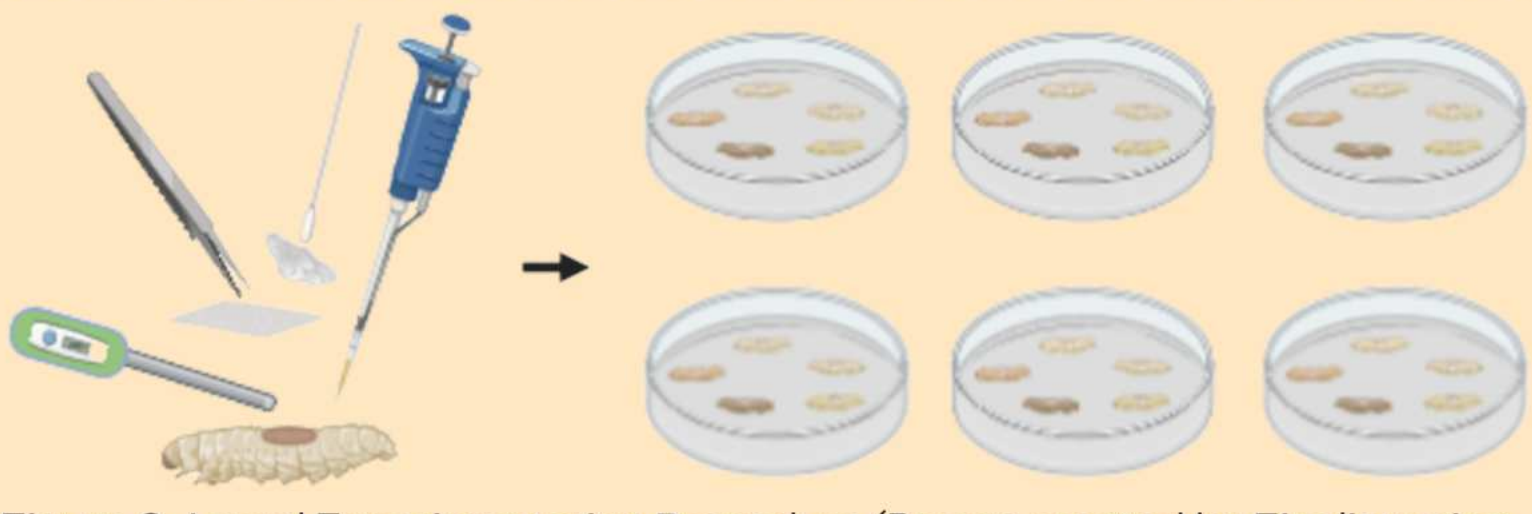


Figure 6. Larval Experimentation Procedure (Image created by Finalist using BioRender)

Larva Burning Procedure

30 larvae for each testing group
300 C, 4 seconds, 10 uL 10% bacteria



Figure 7. Soldering tool used to burn larvae (Photo taken by Finalist)

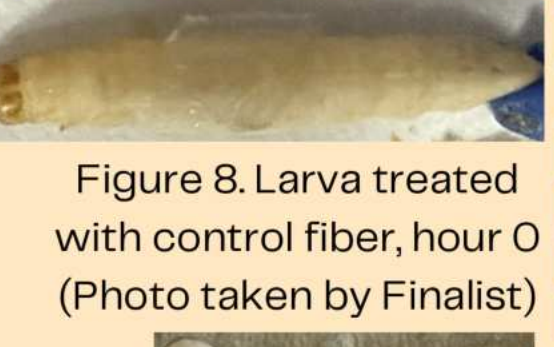


Figure 9. Larva treated with control hydrogel, hour 0 (Photo taken by Finalist)

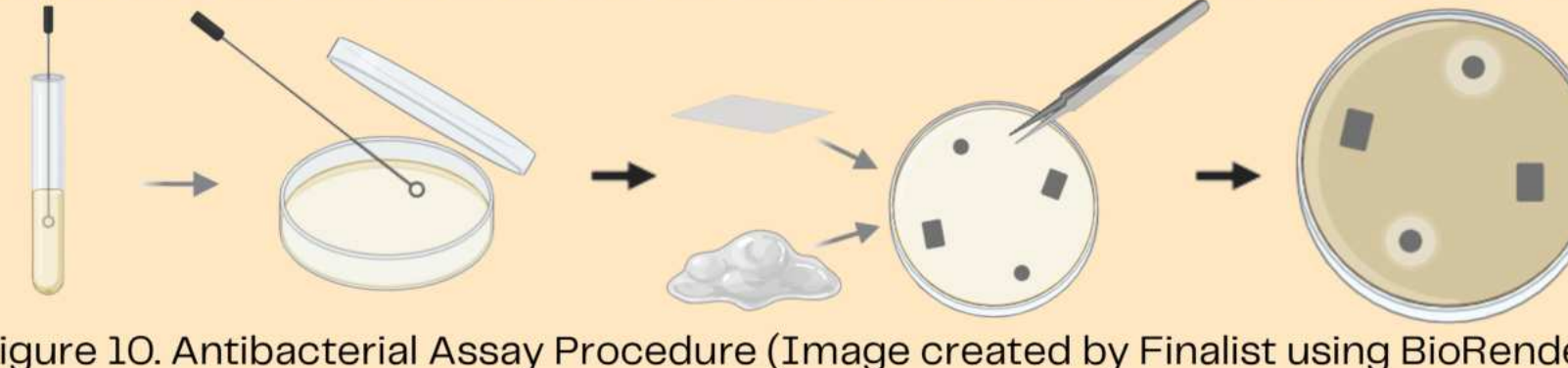


Figure 10. Antibacterial Assay Procedure (Image created by Finalist using BioRender)

Antibacterial Assay

Inoculate *B. cereus* in petri dishes, place treatment 4x, observe differences in growth

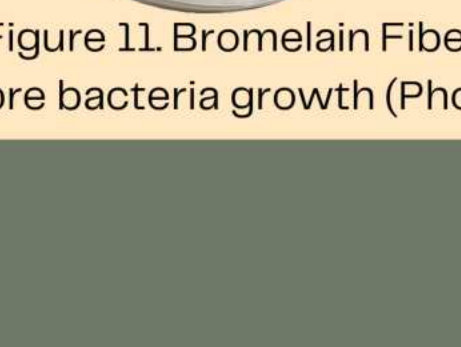


Figure 11. Bromelain Fiber pre bacteria growth (Photo taken by Finalist)

Results

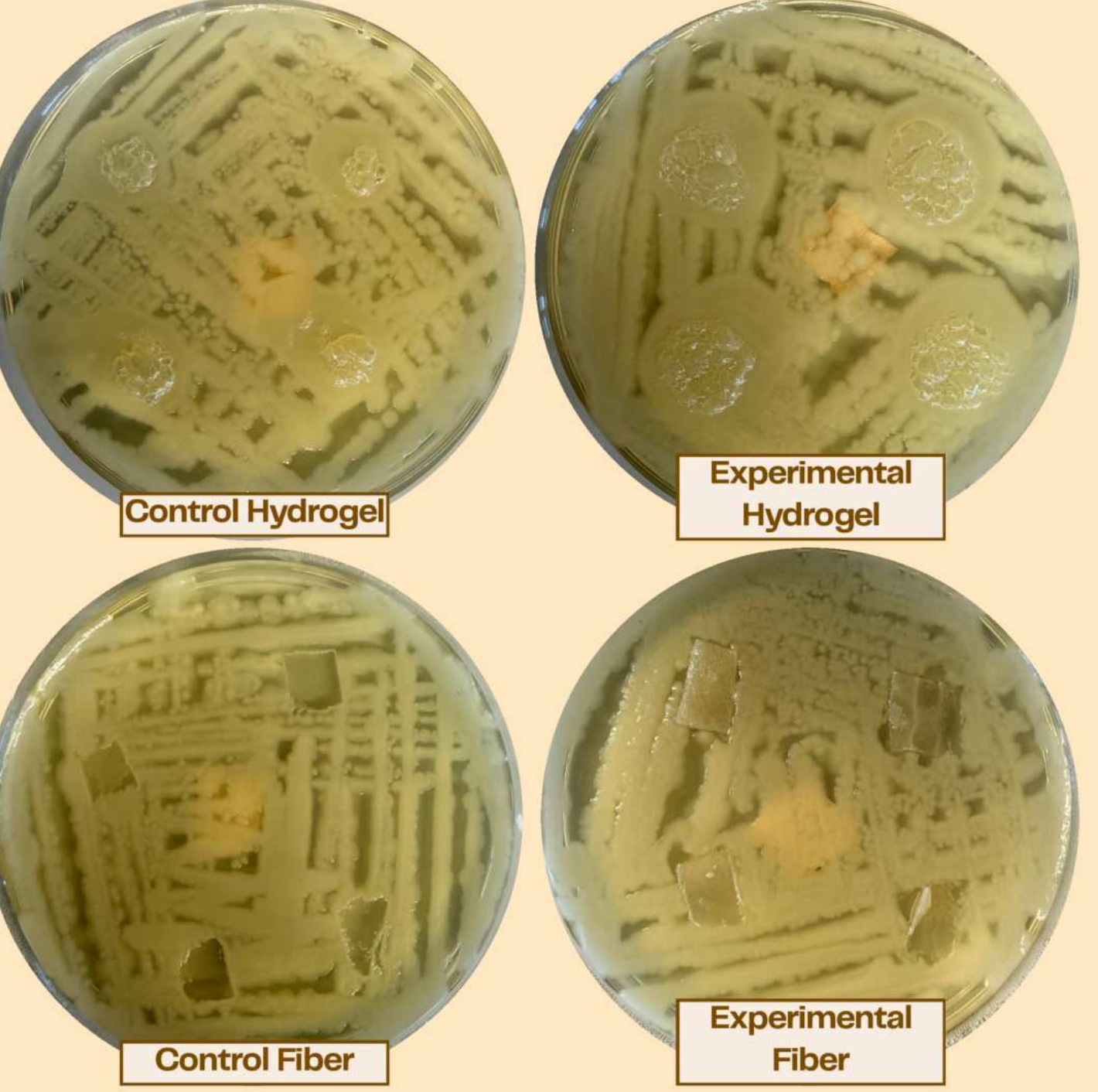


Figure 12. Visual results of antibacterial assay (Photo taken by Finalist)



Figure 13. Larva 12 hour post burn treated with plain chitosan hydrogel (Photo taken by Finalist)



Figure 14. Larva 36 hour post burn treated with plain chitosan electrospun fiber (Photo taken by Finalist)

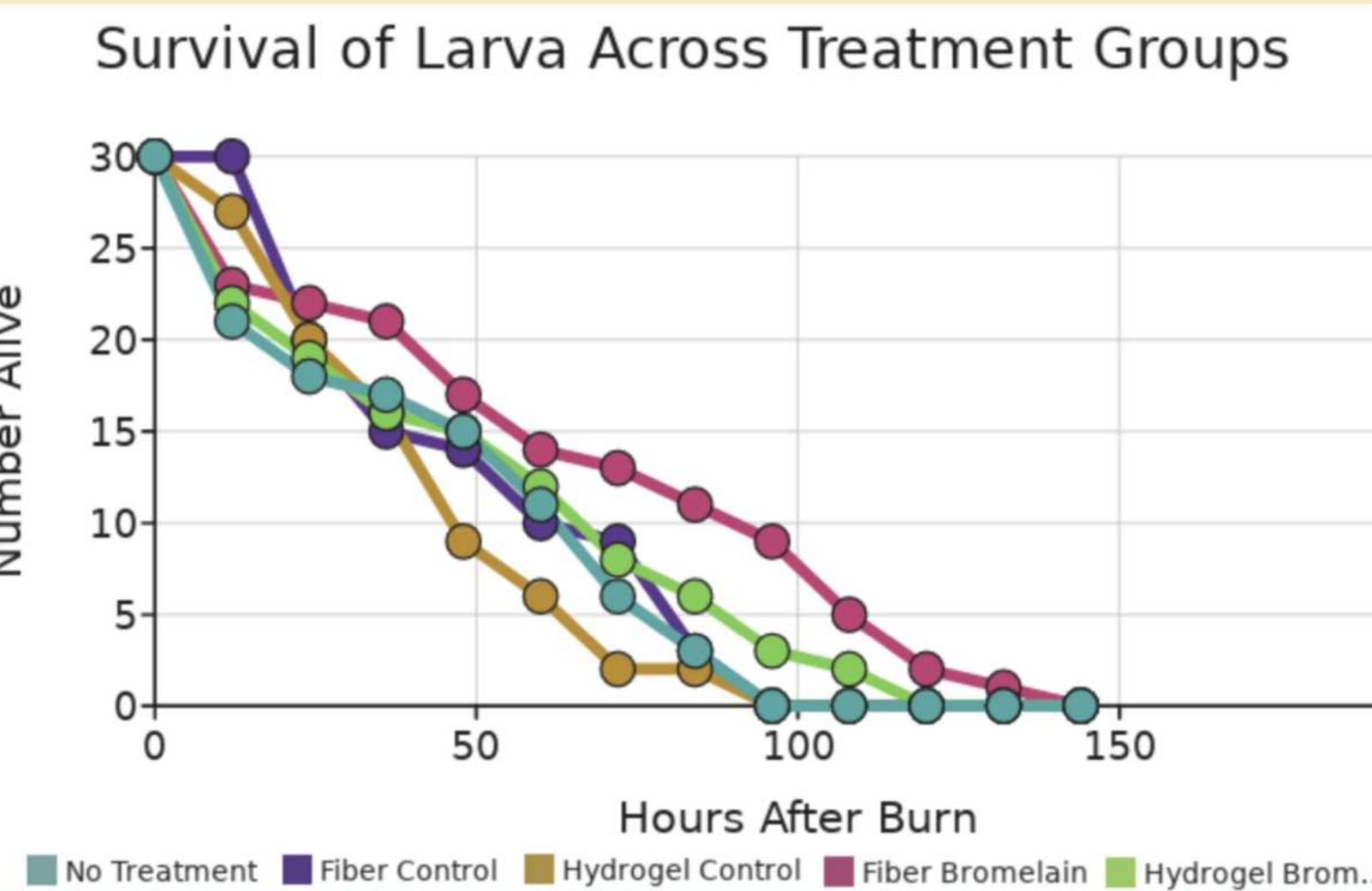


Figure 15. Survival curves of the treatment groups over testing period (Graph created by Finalist using DataClassroom)

Larva given **bromelain** treatments **survived** longer than those treated without it. The **fiber** of bromelain had the most visual **difference** on the graph in terms of **survival**, lasting longer than other treatment groups. The **log-rank** test confirmed the improved survival between experimental and control groups along with the **bromelain fiber** and hydrogel.

Conclusions

Addition of Bromelain

In both the fiber and hydrogel, adding bromelain significantly increased the survival of the larva ($p < 0.05$). Thus, the addition of **bromelain improved** wound healing within the model.

Fiber v. Hydrogel

Without bromelain, there was strong indication, but not statistically significant, of a difference in how the fiber and hydrogel impacted wound healing in the larva ($p = 0.07$). **With bromelain**, there was a significant difference in the survival of the larva between the delivery methods ($p < 0.05$). The **electrospun fiber** prolonged survival, and thus **improved wound healing to greater extents** than the hydrogel.

Galleria Mellonella

Larvae provide a cost-effective, accurate model for first steps of developing new burn treatment

Antibacterial Properties

The fibers both showed **no growth** where placed, while the hydrogels did. This provides qualitative evidence that the **fibers** outperform hydrogels for antibacterial properties.

Future Work

Complete Antibacterial Assay

Confirm **procedural** variables are not causing differences in bacteria **growth**, but rather the form of **treatment**

Sample Size Increase

Test with large sample size ($n > 30$) per treatment to have more **accurate** comparison of the control fiber and control hydrogel

Mammalian Investigation

Use **murine** model to validate efficacy of fiber for burn wound treatment.

Mechanism Study

Examine the **physicochemical properties** of the chitosan and bromelain in **fibers** that allowed for **enhanced wound healing**

Clinical Application Logistics

Determining **how** the fibers can be **applied** for treating wounds on **larger burns** and various **burn types** to more accurately model real-world burn scenarios.

Works Cited

Bayat, S., Amiri, N., Pishavar, E., Kaldinina, F., Movaffagh, J., & Hashemi, M. (2019). Bromelain-loaded chitosan nanofibers prepared by electrospinning method for burn wound healing in animal models. *Life Sciences*, 229, 57–66.

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Xia Y, Wang D, Liu D, Su J, Jin Y, Wang D, Han B, Jiang Z and Liu B (2022) Applications of Chitosan and its Derivatives in Skin and Soft Tissue Diseases. *Front. Bioeng. Biotechnol.* 10:894667. doi: 10.3389/fbioe.2022.894667